

Auteur: Done Goker
Studierichting: Informatica
Oefeningenreeks 2B nr. 1.4

Opgave: $\cos(\text{Bgsin } x)$

Stel $y = \text{Bgsin } x$ met $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$$\Rightarrow \sin y = x$$

$$\Rightarrow \sin^2 y = x^2$$

$$\Rightarrow 1 - \sin^2 y = 1 - x^2$$

$$\Rightarrow \cos^2 y = 1 - x^2$$

$$\Rightarrow \cos y = \sqrt{1 - x^2}$$

$$\Rightarrow \cos(\text{Bgsin } x) = \sqrt{1 - x^2}$$

References